

ECAI for a Multidimensional Network State Directory

Proposal Document

1. Scenario (as raised by Andrew Noble)

The vision is to build a phone book (or yellow pages) for the Network State:

- Records for businesses by jurisdiction and by business category.
- Price information expressed directly in satoshis.
- Businesses and individuals must be able to update their own stale records.
- Records should be usable for tax reasoning, aligned with Logical English.
- A single unit of account for every information repository globally, reducing degrees of freedom in record-keeping.

2. Core Problems

- Traditional directories are centralized and quickly become stale.
- Updates lack verifiability; no immutable proof of who changed what.
- Price denominated in fiat introduces conversion risk and fragmentation.
- Tax and compliance use-cases require deterministic, machine-verifiable records.

3. Proposed Solution with ECAI

Elliptic Curve AI (ECAI) encodes business knowledge deterministically as elliptic curve points, removing guesswork and providing verifiable, immutable state.

3.1 Knowledge Encoding

Each business record is represented in Subject–Predicate–Object–Context structure:

- Subject: Business entity (jurisdiction + name).
- Predicate: Attribute (category, license, address, price, compliance tag).
- Object: Value of the attribute.
- Context: Timestamp and cryptographic signature.

Encoded via ECAI, each record becomes a curve point (knowledge NFT). Updates are signed by the business owner’s keypair, ensuring authenticity.

3.2 Logical English Alignment

Statements like “Company A is licensed in Jurisdiction X” or “Service Y costs 2500 sats” can be written in Logical English and mapped directly into ECAI knowledge points, making them usable for tax reasoning or compliance automation.

3.3 Price in Satoshis

Prices are always denominated in satoshis, binding each record to Bitcoin's native unit. Integration with Lightning invoices allows discovery and settlement in the same structure.

3.4 Multidimensional Queries

Because every record is structured deterministically, queries can be executed across dimensions:

- By jurisdiction (all licensed entities in Jurisdiction X).
- By category (all suppliers in Category Y).
- By compliance status (entities with up-to-date filings).
- By last updated (filtering stale vs. current records).

3.5 Updating Records

Businesses update their own records using private keys. Each new update creates an immutable successor state while preserving the old record for audit history. Consumers and auditors can always verify the most recent valid state.

4. Benefits

- Immutable, cryptographically verifiable records.
- Self-service updates with no central authority required.
- Unified denomination in sats simplifies payments and reporting.
- Compatibility with Logical English enables direct use in tax reasoning.
- Collapses redundant degrees of freedom: one deterministic state layer for global business knowledge.

5. Next Steps

- Build a proof-of-concept registry using DamageBDD + ECAI encoding.
- Define canonical predicates (jurisdiction, category, license, price).
- Integrate Lightning Network payments for record verification and updates.
- Pilot with a small set of businesses across two jurisdictions.